Fine-tuning a Small LLM with LoRA for Automated Paper Review Insights*

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1 Introduction

Peer review remains a foundational pillar of the scientific process, ensuring rigor, validity, and quality through independent expert evaluation (Bornmann, 2011). However, as the number of submissions to journals and conferences continues to grow, the peer review system faces increasing strain, leading to delayed editorial decisions, inconsistent feedback, and reviewer fatigue (Aczel et al., 2025). In parallel, recent advancements in Large Language Models (LLMs) have opened new opportunities for assisting or partially automating aspects of the review process. Current approaches vary in cost, flexibility, and performance, and include: (1) prompt-based zero-shot generation, (2) full supervised fine-tuning on human review data, and (3) parameter-efficient fine-tuning such as Low-Rank Adaptation (LoRA). Prompt engineering involves using general-purpose LLMs (e.g., GPT-4) in a zero-shot setting to generate review-like text given paper abstracts and task-specific instructions (Gilardi et al., 2023). This approach is simple and requires no additional training, but its outputs are highly sensitive to prompt design and lack domain adaptation. In contrast, full supervised fine-tuning retrains all parameters of an LLM on large corpora of human-written reviews, producing highly specialized models (Zhou et al., 2022). While this method can achieve strong task-specific performance, it demands significant computational resources and large-scale annotated datasets. Parameter-efficient fine-tuning techniques, such as LoRA (Hu et al., 2021), offer a compromise by updating only a small number of low-rank parameters within the model, enabling task adaptation even with limited data and compute.

This project investigates whether LoRA fine-tuning can improve the ability of a small open-source LLM—specifically LLaMA 3.2B Instruct model[1]— to extract structured strengths and weaknesses from scientific manuscripts derived from the OpenReview platform [2], and whether such fine-tuning influences the model's ability to estimate numeric review scores. The overarching aim is to build a lightweight, fully reproducible system capable of producing interpretable, structured reviewer-style output drawn from abstracts, titles, and full review text. Our evaluation compares the zero-shot behaviour of the base model against its LoRA-tuned counterpart, using both regression metrics for numeric ratings and semantic measures for the quality of extracted strengths and weaknesses. By examining these two paradigms under controlled conditions, we aim to evaluate whether parameter-efficient fine-tuning constitutes an advantage in low-resource environments or whether prompt conditioning alone is sufficient.

*Project repository on GitHub: `https://github.com/camillabonomo02/Automated_Paper_Review.git`

[1] LLaMa 3.2B Instruct model: `https://huggingface.co/meta-llama/Llama-3.2-3B-Instruct`

[2] OpenReview platform: `https://github.com/Seafoodair/Openreview/tree/master`

2 Methods

The dataset used in this study was assembled from the OpenReview platform, where both peer reviews and associated metadata are publicly accessible. Text fields—including titles, abstracts, and review bodies—were cleaned by removing URLs, normalizing whitespace, and stripping templated markup. Ratings were extracted from heterogeneous formats and converted to a consistent 1–10 numerical scale. The cleaned dataset was divided into training and validation partitions, with care taken to preserve the distribution of scores.

Since the OpenReview dataset does not provide structured annotations for strengths and weaknesses, we adopt a weakly supervised strategy to construct training targets. Specifically, the LLM operating in a zero-shot setting is used as a teacher to generate structured strengths and weaknesses from the paper title and abstract only. These pseudo-labels take the form of concise JSON objects containing a limited number of bullet points for each category.

The student model is then fine-tuned using LoRA to reproduce this structured output. During training, the model receives as input the paper title, abstract, and the full review text is additionally provided as contextual information. Importantly, however, the supervision signal remains entirely derived from the teacher-generated JSON. This design allows the model to leverage the richer linguistic context of the review while still being trained to match a consistent and well-defined output structure.

As a consequence, the fine-tuning objective does not directly optimize alignment with human-written strengths and weaknesses. Instead, it encourages the model to internalize a structured representation of paper evaluation as induced by the teacher model. This approach enables scalable training without manual annotation, but it also introduces an inherent limitation: the quality of the learned representations is bounded by the quality of the pseudo-labels themselves.

In parallel, rating prediction was handled separately. Those are not explicitly learned during fine-tuning. Instead, they are inferred at inference time using a logit-based approach: the model is prompted to output a single scalar value on a 1–10 scale, and the final prediction is computed as the expected value over candidate scores, weighted by their normalized token-level log probabilities.

Because this procedure is sensitive to scale mismatches between model confidence and human scoring distributions, a lightweight post-hoc calibration step is applied. A Huber regressor is fitted on a subset of the training data by mapping predicted scores to ground-truth human ratings. This calibration is applied only during evaluation and does not influence the model's parameters.

Since the LoRA fine-tuning objective does not involve rating supervision, any changes in rating performance between the zero-shot and fine-tuned models should be interpreted as representational side effects of the adaptation process rather than as task-specific learning.

Evaluation proceeded along two complementary dimensions. For rating prediction, traditional regression metrics—MAE, RMSE, and Pearson correlation—were computed before and after calibration to determine whether the model's numeric judgements aligned with human reviewers. For strengths and weaknesses, we measured fallback frequency, assessed lexical diversity across all generated bullets, and computed the cosine similarity between embeddings of teacher-generated and model-generated outputs. A small qualitative inspection was performed to understand failure patterns and identify cases where fine-tuning improved or degraded the informativeness of the extracted content.

3 Results

The two models—zero-shot and fine-tuned—exhibited markedly different behaviour on the numeric rating task (Table 1). The results show that zero-shot prompting already achieves moderate alignment with human ratings, particularly after calibration, while the fine-tuned model does not exhibit systematic improvements and, in some cases, displays degraded correlation with human judgments. This behavior is consistent with the fact that rating prediction is not directly optimized during fine-tuning. Calibration substantially reduces scale-dependent error metrics such as MAE and RMSE for both settings, while correlation metrics remain largely unchanged. This is expected, as calibration primarily corrects for distributional bias rather than rank ordering. In the fine-tuned setting, correlation remains unstable, suggesting that LoRA adaptation on structured pseudo-labels may alter latent representations relevant for scoring without improving their consistency with human preferences.

Task	MAE	RMSE	Pearson
Zero-shot raw	2.72	3.26	0.36
Fine-tuned raw	3.05	3.69	0.07
Zero-shot calibrated	1.82	2.06	0.36
Fine-tuned calibrated	1.99	2.19	-0.07

Table 1: *Regression metrics before and after calibration*

On the structured extraction task, the contrast between the two models was much smaller (Table 2). To assess the quality of extracted strengths and weaknesses, we evaluate structural validity, lexical diversity, and semantic similarity to the teacher outputs. All generated outputs were syntactically valid JSON objects, and explicit structural failures were not observed. However, some outputs became increasingly generic, indicating a tendency toward template-like behavior rather than genuinely diverse critique. Lexical diversity scores show a slight decrease for the fine-tuned model compared to the zero-shot baseline, suggesting a mild homogenization effect induced by fine-tuning. Similarly, semantic similarity to the teacher, measured via cosine similarity in embedding space, remains moderate for both models, with the fine-tuned model not exhibiting improved alignment. These findings indicate that LoRA fine-tuning primarily reinforces consistency with the teacher's output style rather than introducing additional nuance or specificity beyond what is already achievable through zero-shot prompting.

Task	Fallback rate	Lexical div.	Teacher sim. cos.
Zero-shot	0.0	0.23	0.54
Fine-tuned	0.0	0.19	0.51

Table 2: *Semantic evaluation of models*

Overall, the results point toward an asymmetric conclusion: while zero-shot prompting already provides surprisingly strong structured extraction and moderately reliable rating estimation, the LoRA fine-tuned model did not offer meaningful improvements and performed worse on the rating task.

4 Final Discussion

Several avenues for improving fine-tuning performance were considered, including increasing the size of the pseudo-labeled training set and extending the number of training epochs. While such modifications can improve coverage and stability, they do not address a more fundamental limitation of the current approach: the supervision signal is entirely generated by the same class of model used as the student. As a result, fine-tuning primarily increases exposure to the teacher's behavior rather than introducing new information. If the teacher produces strengths and weaknesses that are broadly correct but not particularly insightful, the fine-tuned model can at best learn to replicate that level of quality. Increasing the number of pseudo-labeled examples therefore improves statistical coverage without altering the underlying "informativeness" of the signal. More broadly, this highlights a key limitation of training models to extract strengths and weaknesses in the absence of human-annotated ground truth. The model is optimized to match a synthetic representation of review quality rather than human judgment itself. While this enables scalable experimentation, it also introduces a ceiling effect and limits interpretability: improved similarity to the teacher does not necessarily correspond to improved usefulness for real peer review. These results suggest that, under low-resource conditions, carefully designed zero-shot prompting remains a strong baseline for structured peer-review assistance. Future work may benefit from incorporating even small amounts of human-labeled strengths and weaknesses, or from adopting multi-task objectives that jointly model structured critique and numeric evaluation, thereby reducing representational interference and improving alignment with human reviewers.

5 References

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